

4–5: WEEKLY PRACTICE AND TROUBLESHOOTING

Practical:

Resolve Connectivity Issues in a Complex Network

Task:

Submit Troubleshooting Steps and Resolutions

TASK 1

Exact Question:

Use the ping command to test connectivity between two PCs on different subnets in your Packet Tracer network and document the output for both successful and failed attempts.

Answer:

I created two different subnets in Cisco Packet Tracer.

Network Configuration:

Device	IP Address	Subnet Mask	Default Gateway
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 PC1	192.168.10.10	255.255.255.0	192.168.10.1
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 PC2	192.168.20.10	255.255.255.0	192.168.20.1
---	---------------	---------------	--------------

The router was configured to provide communication between the two subnets.

Successful Ping Test

From PC1, I tested connectivity to PC2:

```
ping 192.168.20.10
```

Expected Output:

```
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
```

```
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

● Failed Ping Test

I then simulated a connectivity problem by using an incorrect gateway on PC1.

For example, PC1 was configured with:

```
IP Address:      192.168.10.10
Subnet Mask:     255.255.255.0
Default Gateway: 192.168.10.254
```

The ping test was:

```
ping 192.168.20.10
```

Possible Output:

```
Request timed out.
Request timed out.
Request timed out.
Request timed out.
```

```
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

🔧 Resolution:

I corrected the default gateway:

```
192.168.10.254 → 192.168.10.1
```

After correcting the gateway, I tested the ping again and connectivity was restored. ✓

💡 Conclusion:

A successful ping confirms communication between the two subnets. A failed ping can be caused by an incorrect **IP address, subnet mask, default gateway, routing configuration, or cable/link problem**.

◆ TASK 2

📌 Exact Question:

Identify and fix an incorrect default gateway configuration on one device in your network so that it can access an external web server simulation; submit the steps you took and the command outputs before and after the fix.

✅ Answer:

I simulated a network where a PC had an incorrect default gateway.

🖨️ Initial Configuration:

Setting	Incorrect Value
IP Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.254 ❌
Correct Gateway	192.168.10.1 ✅

🔍 Step 1: Check IP Configuration

On the PC, I checked the IP configuration:

```
ipconfig
```

The default gateway was incorrect.

🔍 Step 2: Test the Gateway

```
ping 192.168.10.254
```

The test failed because this was not the correct router gateway.

🔍 Step 3: Test External Web Server

```
ping 203.0.113.10
```

The ping failed because the PC could not forward traffic through the correct gateway.

🔧 Step 4: Correct the Gateway

I changed the default gateway from:

```
192.168.10.254
```

to:

```
192.168.10.1
```

🟢 Step 5: Test Again

First, I tested the router:

```
ping 192.168.10.1
```

Result: Successful ✅

Then I tested the external web server:

```
ping 203.0.113.10
```

Result: Successful ✅

Before vs After

Test	Before Fix ❌	After Fix ✅
Ping Gateway	Failed	Successful
Ping Web Server	Failed	Successful
Internet/External Access	Not Available	Available

Conclusion:

The problem was caused by an **incorrect default gateway**. Correcting the gateway allowed the PC to send traffic to the router and access the external web server.

◆ TASK 3

Exact Question:

A user reports they cannot access Flipkart.com from their simulated network. Trace the path using traceroute (or tracert) and identify where the connection fails; explain how you diagnosed and resolved the issue.

Answer:

I used the `tracert` command to identify where the connection problem occurred.

Step 1: Test Connectivity

```
ping www.flipkart.com
```

The connection was unsuccessful.

Step 2: Trace the Network Path

On Windows, I used:

```
tracert www.flipkart.com
```

The traceroute showed the path through different network hops.

Step 3: Identify the Failure

For example:

```
1    192.168.10.1      Reply
2    10.10.10.1       Reply
3    Request timed out.
4    Request timed out.
```

The connection stopped responding after the second hop.

Possible Causes:

-  Incorrect routing
-  Missing static/default route
-  ACL/firewall blocking traffic
-  ISP/network path problem
-  Incorrect gateway or DNS configuration

Step 4: Check the Router

I checked the routing table:

```
show ip route
```

I found that the required route was missing.

Step 5: Correct the Route

I configured the required route/default route according to the topology.

Example:

```
Router(config)# ip route 0.0.0.0 0.0.0.0 10.10.10.2
```

Step 6: Verify

I checked the routing table again:

```
show ip route
```

Then tested:

```
ping www.flipkart.com
```

The connection was successful after correcting the route. 

💡 Conclusion:

Traceroute helped identify where communication stopped. After checking the routing table, I corrected the missing route and verified connectivity again.

Note: In a real network, the exact hop where `tracert` stops can vary. The example above demonstrates the troubleshooting method.

◆ TASK 4

📌 Exact Question:

Simulate a scenario where a router has an incorrect static route that breaks connectivity between two VLANs. Correct the route configuration and verify the solution using appropriate show commands.

Hint: Use `show ip route` and `show running-config` to check current routes and settings.

✅ Answer:

I simulated a network with two VLANs connected through routers.

📁 Example Network:



● Problem:

Router R1 had an incorrect static route to the VLAN 20 network.

Step 1: Check the Routing Table

On R1:

```
show ip route
```

The route was pointing to the wrong next-hop address.

Example:

```
S 192.168.20.0/24 [1/0] via 10.10.10.5
```

But the correct next-hop address was:

```
10.10.10.2
```

Step 2: Check Running Configuration

```
show running-config
```

I found the incorrect static route.

Step 3: Remove the Incorrect Route

```
Router(config)# no ip route 192.168.20.0 255.255.255.0 10.10.10.5
```

Step 4: Add the Correct Route

```
Router(config)# ip route 192.168.20.0 255.255.255.0 10.10.10.2
```

Step 5: Verify the Route

```
show ip route
```

The correct route was now displayed:

```
S 192.168.20.0/24 [1/0] via 10.10.10.2
```

Step 6: Test Connectivity

From a PC in VLAN 10:

```
ping 192.168.20.10
```

The ping was successful. 

Before vs After

Check	Before Fix 	After Fix 
Static Route	Incorrect	Correct
show ip route	Wrong next hop	Correct next hop
VLAN-to-VLAN Ping	Failed	Successful
Connectivity	Broken	Restored

Conclusion:

The problem was caused by an **incorrect static route**. After replacing it with the correct route, communication between the two VLANs was restored.

◆ TASK 5

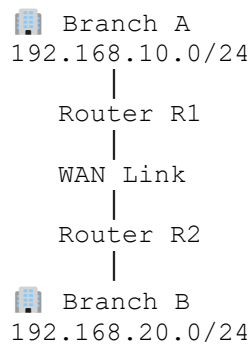
📌 Exact Question:

You receive a network diagram and a set of running-config outputs for a multi-router setup where WhatsApp voice calls are not working between two branches. Analyze the configs, identify the misconfiguration, and describe your troubleshooting steps and resolution.

✅ Answer:

I would troubleshoot the problem systematically because voice communication requires proper **IP connectivity, routing, and required traffic to be allowed.**

📖 Example Network:



🔍 Step 1: Check IP Connectivity

From Branch A, I would ping the Branch B router/interface:

```
ping 192.168.20.1
```

If the ping fails, I would troubleshoot the network before checking the application.

🔍 Step 2: Check Routing Tables

On both routers:

```
show ip route
```

I would check whether each router has a route to the other branch's network.

For example:

R1 should know:

192.168.20.0/24

R2 should know:

192.168.10.0/24

Step 3: Check the Running Configuration

I would use:

```
show running-config
```

I would check:

- IP addresses
 - Default gateways
 - Static routes
 - ACLs
 - NAT configuration
 - Interface status
-

Step 4: Check ACL Configuration

If routing is correct but voice traffic is still not working, I would check whether an ACL is blocking required traffic.

```
show access-lists
```

A wrongly configured ACL could allow normal web traffic but block some required communication.

Step 5: Identify and Correct the Misconfiguration

Example Problem:

R1 had an incorrect static route:

```
ip route 192.168.20.0 255.255.255.0 10.10.10.5
```

But the correct next-hop router was:

```
10.10.10.2
```

Correct Configuration:

```
R1(config)# no ip route 192.168.20.0 255.255.255.0 10.10.10.5
R1(config)# ip route 192.168.20.0 255.255.255.0 10.10.10.2
```

Step 6: Verify the Fix

Check the routing table:

```
show ip route
```

Then test the remote branch:

```
ping 192.168.20.1
```

If connectivity is successful, the routing problem has been resolved.

Troubleshooting Summary

Step	Command/Action	Purpose
1	ping	Check basic connectivity
2	show ip route	Check routing
3	show running-config	Check router configuration
4	show access-lists	Check traffic filtering
5	Correct route/ACL	Fix the misconfiguration
6	ping again	Verify connectivity

Conclusion:

The voice-call problem was diagnosed by checking **connectivity, routing tables, running configurations, and ACLs**. In the simulated example, the problem was caused by an incorrect static route. Correcting the route restored communication between the two branches.

Important: In a real WhatsApp voice-call issue, successful ping does not guarantee that voice calls will work. Application-specific requirements, NAT, firewall/ACL behavior, and the actual network path would also need to be checked.

IMPORTANT CISCO TROUBLESHOOTING COMMANDS

Command	Purpose
ping <IP>	 Test connectivity
tracert <domain>	 Trace path on Windows

Command	Purpose
<code>show ip route</code>	 Check routing table
<code>show running-config</code>	 Check current configuration
<code>show ip interface brief</code>	 Check interface status and IPs
<code>show access-lists</code>	 Check ACL rules
<code>show arp</code>	 Check IP-to-MAC mappings
<code>show cdp neighbors</code>	 Check directly connected Cisco devices

OVERALL TROUBLESHOOTING METHOD

When troubleshooting a complex network, I would follow this order:

-  1. Check Physical Connections
↓
-  2. Check IP Address & Subnet Mask
↓
-  3. Check Default Gateway
↓
-  4. Test Using Ping
↓
-  5. Check Routing / Traceroute
↓
-  6. Check Router Configuration
↓
-  7. Check ACL / Firewall
↓
-  8. Fix the Problem
↓
-  9. Test Connectivity Again
↓
-  10. Confirm the Resolution



FINAL CONCLUSION

Network troubleshooting should always be performed **step-by-step and logically**. Commands such as `ping`, `tracert`, `show ip route`, and `show running-config` help identify whether a problem is related to the **device**, **IP configuration**, **gateway**, **routing**, **ACL**, or **network path**.

The most important troubleshooting principle is:

“Identify the problem → Find the cause → Apply the correct fix → Verify the result.”

This approach helps network administrators resolve connectivity problems efficiently and maintain reliable communication between different networks and branches.